

Drug Safety Signal Detector

& Regulatory Submission Readiness Checker

Team TechTress | IBM Bob AI Innovation Hackathon 2026 | AI Track

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Local Streamlit prototype — not a certified PV or CTD compliance system

Problem

FAERS-scale adverse-event volume makes manual signal hunting unreliable.

ICH M4 CTD dossiers span five modules; missing sections delay filings.

Reviewers need reproducible PRR screening and a representative gap list.

Solution — two modes

1. Signal Detection: clean CSV → patterns → 2x2 PRR → rank potential signals.
2. Submission Readiness: map outline → five CTD modules → score → gap report.

Rule-based explanations. No external AI API at runtime.

Architecture

Reviewer → Streamlit → Pandas/NumPy → scikit-learn (optional clustering)

→ PRR engine / CTD checker → Plotly + CSV export

IBM Bob: development workflow only (planning, code, tests, docs).

PRR screening

$$\text{PRR} = [a / (a + b)] / [c / (c + d)]$$

Handles empty margins and division by zero.

Flags 'Potential Safety Signal' — not a medical conclusion.

CTD readiness

Representative checklist across Modules 1-5.

Module completeness, weighted overall score, High/Medium/Low gaps.

Downloadable report: Module | Section | Status | Priority | Recommendation

Limitations

Prototype only. Sample/synthetic data is not complete FAERS.

Not a replacement for pharmacovigilance experts.

CTD list is representative, not exhaustive.

Team TechTress

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Run: `python -m streamlit run src/app.py`